MYCO AI – mushroom phase reference cards

Printable demo set · built 2026-10-16 · 3 species · 13 cards

How the demo works

1. Print this PDF (A4, colour if possible) and cut out the cards you want to demonstrate.
2. Hold a card flat in front of the ESP32-CAM, 20–40 cm away, filling most of the frame, without glare.
3. Press “Take a photo now” on the MYCO AI Overview page (or wait for the automatic interval).
4. The backend recognises the card – first by its QR code, otherwise by matching the picture’s features – and sets the chamber’s batch to the matching growth phase.
5. The deterministic controller then drives fan, humidifier and bulb towards that phase’s ranges below.

The recogniser only knows these printed cards. It is not a general mushroom classifier.

Phases and the ranges the controller applies

Species	Phase (card)	Profile stage	Temperature	Humidity	Fresh air
Oyster	Colonising	Incubation & colonisation	20–24 °C	60–80 % RH	passive
	Pinning	Primordia formation (pinni	15–20 °C	90–95 % RH	5 min / 30 min
	Fruiting	Fruiting-body development	16–21 °C	85–95 % RH	5 min / 20 min
	Harvest ready	Harvest readiness	16–21 °C	85–95 % RH	5 min / 20 min
Shiitake	Colonising	Incubation & colonisation	20–24 °C	60–75 % RH	passive
	Browning	Browning (block maturation	20–24 °C	60–75 % RH	5 min / 60 min
	Pinning	Pinning (after soak/shock)	12–18 °C	85–95 % RH	5 min / 30 min
	Fruiting	Fruiting-body development	14–20 °C	80–90 % RH	5 min / 30 min
	Harvest ready	Harvest readiness	14–20 °C	80–90 % RH	5 min / 30 min
Lion's mane	Colonising	Incubation & colonisation	20–24 °C	60–75 % RH	passive
	Pinning	Primordia formation	18–22 °C	90–95 % RH	5 min / 30 min
	Fruiting	Fruiting-body development	18–22 °C	85–95 % RH	5 min / 20 min
	Harvest ready	Harvest readiness	18–22 °C	85–95 % RH	5 min / 20 min

Typical ranges compiled from general cultivation literature. Not validated for your strain, substrate or chamber – verify and adjust. Note: the DHT11 is only rated to ~90 % RH and 0–50 °C; readings above 90 % RH should be treated as approximate. Humidity targets are capped at 95 % because that is the DHT11's usable limit; the 12 V fan cannot cool below room temperature, so temperatures above a phase's maximum are reported, not fixed.

Images still missing for: lions_mane_pinning, lions_mane_fruiting, lions_mane_harvest_ready – those pages show a placeholder.

QR payload format: MYCO:<species code>:<phase>. OYS = Oyster, SHI = Shiitake, LMN = Lion's mane.

Oyster · Colonising

Pleurotus ostreatus



What you see

White mycelium spreading through the substrate; no fruit bodies.

- White mycelium spreading from spawn points
- Substrate consolidating into a block

Controller targets for this phase (profile stage “Incubation & colonisation”)

20–24 °C **60–80 % RH** fresh air: passive

Typical duration 10–21 days · next: Substrate fully colonised (white throughout), no green/black patches.
Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:OYS:colonising

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Oyster · Pinning

Pleurotus ostreatus



What you see

First tiny primordia (pins) at the openings.

- Tiny pin clusters at openings
- Mycelium knots on surface

Controller targets for this phase (profile stage “Primordia formation (pinning)”

15–20 °C **90–95 % RH** fresh air: 5 min every 30 min

Typical duration 3–7 days · next: Visible pin clusters at the fruiting openings (grower confirmation).
Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:OYS:pinning

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Oyster · Fruiting

Pleurotus ostreatus



What you see

Young fruit bodies growing, caps still small / domed.

- Caps expanding daily
- Stems short and caps broad with good airflow

Controller targets for this phase (profile stage “Fruiting-body development”)

16–21 °C **85–95 % RH** fresh air: 5 min every 20 min

Typical duration 4–7 days · next: Cap edges begin to flatten; largest caps near harvest size.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:OYS:fruiting

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Oyster · Harvest ready

Pleurotus ostreatus



What you see

Mature fruit bodies at harvest size.

- Cap edges still slightly curled downward
- Cluster firm

Controller targets for this phase (profile stage “Harvest readiness”)

16–21 °C **85–95 % RH** fresh air: 5 min every 20 min

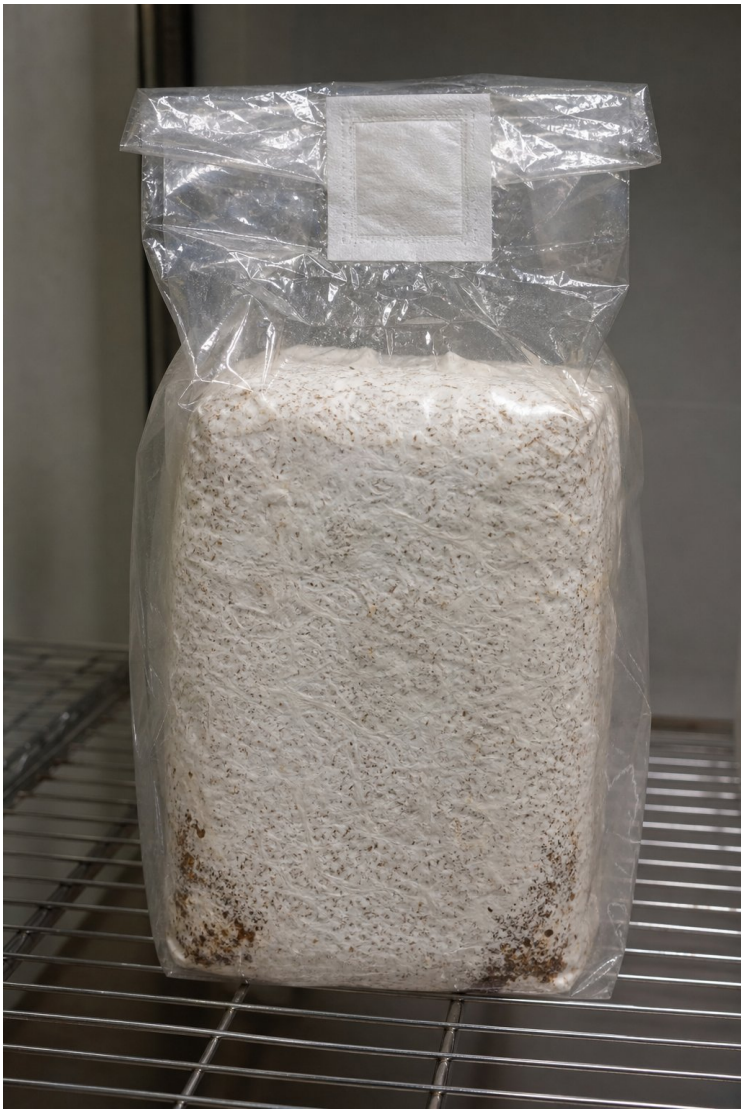
Typical duration 1–2 days · next: Grower judges cluster ready and harvests.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:OYS:harvest_ready

Shiitake · Colonising

Lentinula edodes



What you see

White mycelium spreading through the substrate; no fruit bodies.

• White mycelium through block

Controller targets for this phase (profile stage “Incubation & colonisation”)

20–24 °C **60–75 % RH** fresh air: passive

Typical duration 21–40 days · next: Block fully white (colonised).

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:SHI:colonising

Shiitake · Browning

Lentinula edodes



What you see

Block has left the bag and formed a brown protective skin; popcorning bumps.

- Surface turning brown/tan
- Bumps (popcorning) on the surface

Controller targets for this phase (profile stage “Browning (block maturation)”)

20–24 °C **60–75 % RH** fresh air: 5 min every 60 min

Typical duration 14–35 days · next: Block mostly brown and firm; ready for cold-water soak / shock.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:SHI:browning

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Shiitake · Pinning

Lentinula edodes



What you see

First tiny primordia (pins) at the openings.

• Small brown pins on block surface

Controller targets for this phase (profile stage “Pinning (after soak/shock)”)

12–18 °C **85–95 % RH** fresh air: 5 min every 30 min

Typical duration 5–10 days · next: Pins visible (grower confirmation).

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:SHI:pinning

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Shiitake · Fruiting

Lentinula edodes



What you see

Young fruit bodies growing, caps still small / domed.

- Caps expanding
- Veil stretching

Controller targets for this phase (profile stage “Fruiting-body development”)

14–20 °C **80–90 % RH** fresh air: 5 min every 30 min

Typical duration 7–10 days · next: Veil just breaking on most caps.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:SHI:fruiting

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Shiitake · Harvest ready

Lentinula edodes



What you see

Mature fruit bodies at harvest size.

Controller targets for this phase (profile stage “Harvest readiness”)

14–20 °C **80–90 % RH** fresh air: 5 min every 30 min

Typical duration 1–3 days · next: Grower judges caps ready and harvests.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:SHI:harvest_ready

Lion's mane · Colonising

Hericium erinaceus



What you see

White mycelium spreading through the substrate; no fruit bodies.

- Faint white growth
- Substrate smells sweet, not sour

Controller targets for this phase (profile stage “Incubation & colonisation”)

20–24 °C **60–75 % RH** fresh air: passive

Typical duration 12–21 days · next: Fully colonised.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:LMN:colonising

MYCO AI demo reference card – recognised by QR code or by matching this picture's features. Ranges from the built-in reference profile; verify for your strain.

Lion's mane · Pinning

Hericium erinaceus



What you see

First tiny primordia (pins) at the openings.

• White knots forming at openings

Controller targets for this phase (profile stage “Primordia formation”)

18–22 °C **90–95 % RH** fresh air: 5 min every 30 min

Typical duration 3–7 days · next: Pins visible (grower confirmation).

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:LMN:pinning

Lion's mane · Fruiting

Hericium erinaceus



What you see

Young fruit bodies growing, caps still small / domed.

- Spines elongating
- Fruit body pure white

Controller targets for this phase (profile stage “Fruiting-body development”)

18–22 °C **85–95 % RH** fresh air: 5 min every 20 min

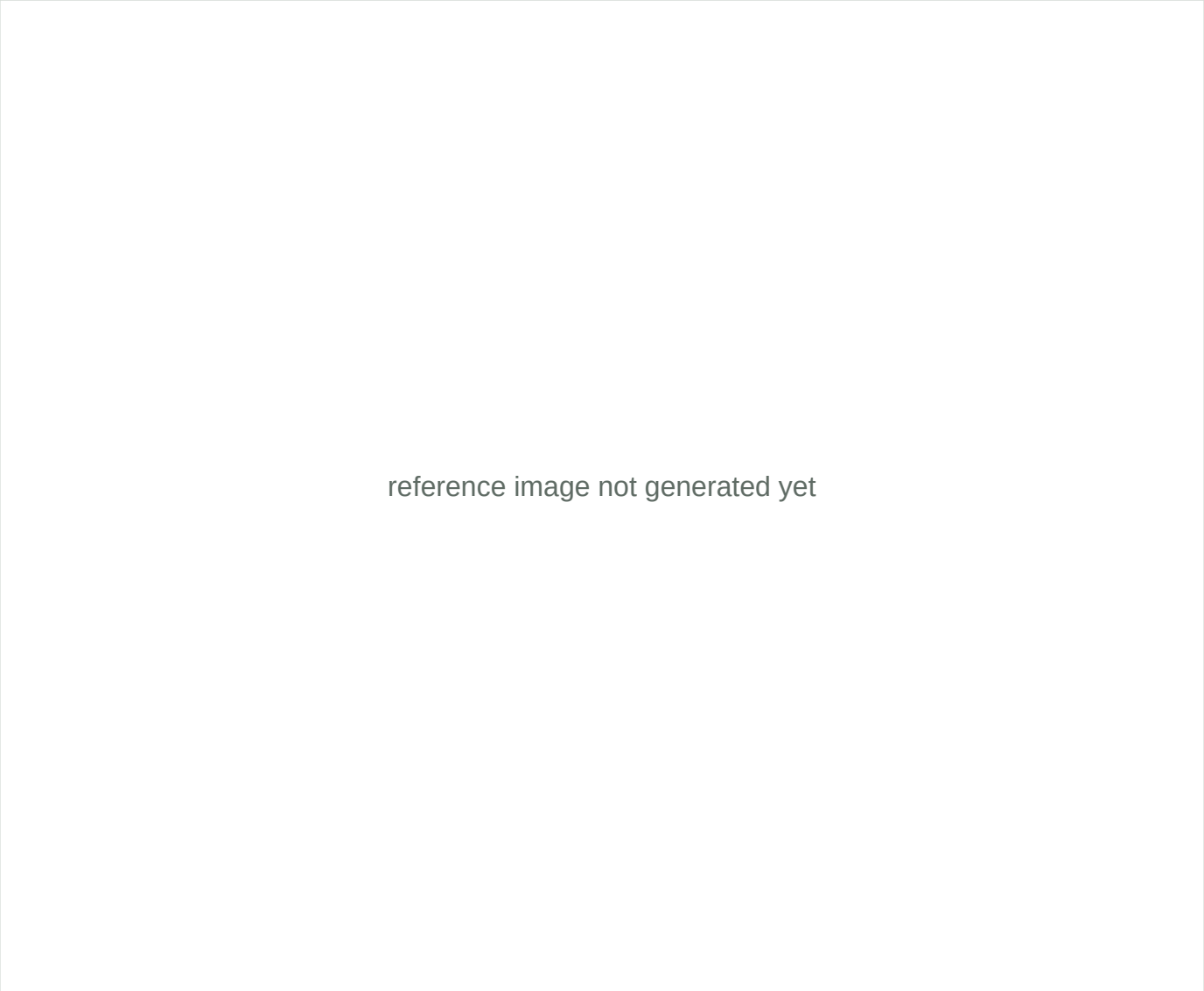
Typical duration 7–14 days · next: Spines approx. 0.5–1 cm long.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:LMN:fruiting

Lion's mane · Harvest ready

Hericium erinaceus



What you see

Mature fruit bodies at harvest size.

Controller targets for this phase (profile stage “Harvest readiness”)

18–22 °C **85–95 % RH** fresh air: 5 min every 20 min

Typical duration 1–3 days · next: Grower judges fruit body ready and harvests.

Humidifier raises RH, fan lowers RH / exchanges air, bulb (relay) heats – all with hysteresis and interlocks.

MYCO:LMN:harvest_ready